

Rates as Prices of Prices

A Bayesian Causal Theory of Higher-Order Economic Dynamics and Regime Change

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Abstract

This paper develops a unified framework in which prices, rates, rates of rates, and regime changes are interpreted as successive layers of economic information. We begin from the observation that many economically important rates are themselves constructed from prices and can therefore be interpreted as prices of prices. Rates of rates emerge naturally as second-order economic quantities analogous to acceleration in physics. We then develop a Bayesian causal framework in which regime changes correspond to changes in latent causal structures rather than merely changes in statistical parameters. The resulting framework links economics, finance, political science, geopolitics, warfare economics, artificial intelligence, statistics, and computer science through a common hierarchy of economic dynamics.

The paper ends with “The End”

Contents

1	Introduction	1
2	A Hierarchy of Economic Objects	1
3	Rates as Prices of Prices	2
3.1	Interest Rates	2
3.2	Inflation	2
3.3	Exchange Rates	2
4	Rates of Rates	2
5	A Theorem on Early Warning Signals	3
6	Bayesian Regime Inference	3
7	Bayesian Change-Point Detection	4
8	Causal Regime Change	4
9	Political Science and Geopolitical Applications	4
10	Warfare Economics	5
11	Financial Markets	5
12	Machine Learning Interpretation	5
13	Algorithm	6
14	Data Science Architecture	6
15	The Structural-Break-First Principle	6

1 Introduction

Economic systems contain multiple layers of information.

At the lowest level are physical goods and services.

At the next level are prices.

Above prices are rates.

Above rates are rates of rates.

The central thesis of this paper is that rates should be interpreted as prices of prices and that rates of rates provide early information regarding structural transitions, regime shifts, crises, wars, financial panics, and geopolitical realignments.

The framework naturally leads to Bayesian causal inference as a methodology for detecting latent regime changes.

2 A Hierarchy of Economic Objects

Definition 1. *An economic hierarchy is a sequence of increasingly abstract objects generated from underlying economic quantities.*

The proposed hierarchy is:

Level	Object
0	Goods and Services
1	Prices
2	Rates
3	Rates of Rates
4	Rates of Rates of Rates

This hierarchy resembles the hierarchy found in classical mechanics:

Mechanics	Economics
Position	Price
Velocity	Rate
Acceleration	Rate of Rate
Jerk	Rate of Rate of Rate

3 Rates as Prices of Prices

Let

$$P_t$$

denote a price.

A continuously compounded rate is

$$r_t = \frac{d}{dt} \ln(P_t).$$

Since the logarithmic derivative compares one price with another nearby price, the rate itself is generated from prices.

3.1 Interest Rates

Let

$$PV = \frac{FV}{1+r}.$$

Then

$$1+r = \frac{FV}{PV}.$$

The interest rate measures the relative price of future money compared with present money. Thus interest rates are prices governing intertemporal prices.

3.2 Inflation

Inflation is

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

Inflation is therefore a rate constructed from prices.

3.3 Exchange Rates

Let

$$E_{ij} = \frac{\text{Currency } i}{\text{Currency } j}.$$

An exchange rate is the price of one pricing instrument measured in another pricing instrument.

4 Rates of Rates

Define

$$a_t = \frac{dr_t}{dt}.$$

The quantity a_t is a rate of a rate.

Definition 2. *A rate of rate is a second-order economic quantity measuring the acceleration or deceleration of a rate process.*

Examples include:

1. Inflation acceleration.
2. Interest-rate acceleration.
3. Exchange-rate acceleration.
4. Credit-growth acceleration.
5. Military expenditure acceleration.

5 A Theorem on Early Warning Signals

Theorem 1. *Suppose an economic variable follows*

$$P_t.$$

Let

$$r_t = \frac{d}{dt} \ln(P_t)$$

and

$$a_t = \frac{dr_t}{dt}.$$

If a regime transition begins through gradual changes in the growth mechanism before producing observable level effects, then changes in a_t must precede changes in r_t and P_t .

Proof. Observe that

$$P_t \rightarrow r_t \rightarrow a_t$$

forms a hierarchy of derivatives.

Suppose the underlying mechanism changes continuously.

Then the earliest observable effect appears in the highest derivative.

Since

$$a_t = \frac{d^2}{dt^2} \ln(P_t),$$

changes in acceleration necessarily occur before large deviations emerge in rates or levels.

Therefore second-order quantities contain leading information regarding regime transitions. □

6 Bayesian Regime Inference

Suppose

$$Y_t \sim P(Y_t|\theta_t).$$

Traditional models assume

$$\theta_t = \theta.$$

We instead introduce a latent regime variable

$$S_t \in \{1, \dots, K\}.$$

The observation equation becomes

$$Y_t \sim P(Y_t|S_t, \theta_{S_t}).$$

The inference problem is

$$P(S_t|Y_{1:t}).$$

A regime shift occurs when posterior probability migrates from one state to another.

7 Bayesian Change-Point Detection

Define run length

$$R_t.$$

The run length counts observations since the last regime change.
The recursive posterior is

$$P(R_t|Y_{1:t}).$$

A change point occurs when

$$P(R_t = 0|Y_{1:t})$$

becomes large.

This permits online detection of crises and structural breaks.

8 Causal Regime Change

Consider a causal graph

$$G_t = (V_t, E_t).$$

A regime shift occurs whenever

$$G_t \neq G_{t-1}.$$

Three possibilities exist:

1. New causal edges appear.
2. Existing causal edges disappear.
3. Structural parameters change.

The Bayesian problem becomes

$$P(G_t \neq G_{t-1}|D_t).$$

This transforms regime detection from a purely statistical exercise into a causal inference problem.

9 Political Science and Geopolitical Applications

Political systems frequently exhibit hidden regimes.

Examples include:

1. Peace and war.
2. Democracy and autocracy.
3. Globalization and deglobalization.
4. Monetary stability and inflationary breakdown.
5. Alliance formation and alliance collapse.

Let

$$S_t = \{\text{Peace}, \text{War}\}.$$

Bayesian updating provides

$$P(S_t = \text{War}|D_t).$$

Increasing posterior probability serves as an early warning indicator.

10 Warfare Economics

Military expenditure may be represented by

$$M_t.$$

The military growth rate is

$$m_t = \frac{d}{dt} \ln(M_t).$$

The acceleration is

$$\alpha_t = \frac{dm_t}{dt}.$$

Historically, major conflicts are often preceded by increases in α_t before substantial increases become visible in M_t itself.

Thus rates of rates provide strategic intelligence.

11 Financial Markets

Let

$$B(r)$$

be a bond price.

Duration is

$$D = -\frac{1}{B} \frac{dB}{dr}.$$

Convexity is

$$C = \frac{1}{B} \frac{d^2 B}{dr^2}.$$

Convexity is a second-order response to changes in rates.

Therefore modern fixed-income theory already contains implicit rates-of-rates analysis.

12 Machine Learning Interpretation

Let

$$x_t = (P_t, r_t, a_t).$$

A machine learning model receives the feature vector

$$x_t.$$

The objective is

$$P(S_t|x_t).$$

Possible models include:

1. Hidden Markov Models.
2. Bayesian Networks.
3. Particle Filters.
4. Recurrent Neural Networks.
5. Transformers.
6. Reinforcement Learning Agents.

13 Algorithm

Pseudo-Code

```
Input:
Economic observations  $Y$ 

Initialize:
Prior over regimes  $P(S)$ 

For each time period  $t$ :

    Observe  $Y_t$ 

    Compute:
        Price level
        Rate
        Rate of rate

    Update:
         $P(S_t \mid Y_1, \dots, Y_t)$ 

    If  $P(\text{Change Point}) > \text{Threshold}$ :

        Trigger Warning

Output:
Regime probabilities
Early-warning indicators
```

14 Data Science Architecture



15 The Structural-Break-First Principle

Traditional analysis follows:

Parameters $\rightarrow$ Forecasts $\rightarrow$ Break Detection.

The proposed methodology reverses the sequence:

Regime Detection $\rightarrow$ Causal Structure $\rightarrow$ Parameters $\rightarrow$ Forecasts.

This approach places structural change at the center of economic analysis.

16 Conclusion

Rates may be interpreted as prices of prices.

Rates of rates emerge as higher-order economic quantities analogous to acceleration.

Because structural transitions first appear in higher derivatives, rates of rates frequently contain leading information regarding crises, wars, financial instability, inflationary transitions, and geopolitical realignments.

Bayesian causal inference provides a coherent framework for detecting these transitions by estimating latent regimes and evolving causal structures. The resulting synthesis unifies economics, finance, political science, international relations, warfare economics, statistics, computer science, machine learning, and data science within a common theory of higher-order economic dynamics.

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Glossary

Price

Market valuation of a good, service, asset, or claim.

Rate

A price-derived quantity measuring relative change.

Rate of Rate

Acceleration of a rate process.

Regime

A latent state governing data generation.

Bayesian Inference

Probability updating using observed evidence.

Causal Graph

A graph representing causal relationships.

Change Point

A point at which the underlying regime changes.

Structural Break

A discontinuity in the data-generating process.

Warfare Economics

The study of economic systems under military competition and conflict.

The End